



Concussion and Brain Injury

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Among the wide range of injuries that occur in the sports arena, traumatic brain injury (TBI) is a health problem that has garnered increasing public awareness during recent years. This heightened interest has been largely driven by intense media exposure, as well as the retirement of prominent professional athletes with cognitive and behavioral dysfunction purportedly representing lingering sequelae of brain injury. Player safety concerns have become an area of special focus for professional leagues, such as the National Football League (NFL) and National Hockey League (NHL), as well as organizations such as the International Olympic Committee and the National Collegiate Athletic Association (NCAA), with a resulting impact on collegiate, high school, and recreational sports.

Key to a discussion of sports brain injury is an understanding of the terminology applied to the topic. *TBI* is an umbrella term that encompasses a range of clinical labels that vary based on severity. Mild traumatic brain injury (mTBI) involves an impact to or acceleration/deceleration of the head resulting in at least a temporary alteration in consciousness or loss of consciousness (LOC) of less than 20 minutes, a Glasgow Coma Scale score of 13 to 15, and no findings on neuroimaging.¹ mTBI is alternately referred to as *mild head injury*, *minor head injury*, and *concussion*. The vast majority of brain injuries sustained by athletes are mild events that would be considered concussions²; moderate, severe, and penetrating traumatic brain injuries are less common. For practical purposes, in this chapter mTBI, mild head injury, and concussion are considered synonymous, and the term *concussion* is used throughout this chapter.

The intent of this chapter is to provide a resource for the clinician on the diagnosis and management of sports concussion. A brief historical context for the issue of sports concussion, relevant definitions, and epidemiology are presented; the clinical presentations that can occur after a concussive blow are also reviewed, and a clear concussion management and return-to-play (RTP) decision-making process is given.

BACKGROUND

Concussion is not a modern phenomenon. Numerous references to cranial injury can be found in ancient medical reports and mythological literature by such figures as Hippocrates and Homer. The recognition of concussion as a separate entity from more severe head injuries was made as far back as the first century AD by Rhazes, an Arabian physician who described concussion as an

abnormal physiologic state lacking gross traumatic brain lesions. Modern-day literature, however, has predominately focused on moderate to severe brain injuries, with relatively little attention given to mild brain injuries, whether sports-related or not.

The topic of concussion began to be taken seriously by medical science in the 1980s, when clinical and epidemiologic studies received media attention in a *Wall Street Journal* article, “Silent Epidemic: Head Injuries Often Difficult to Diagnose, Get Rising Attention.” The growing consensus that mild head injury was in fact not innocuous was based on the identification of neuropsychological deficits in problem solving, attention, and memory lasting up to 3 months after a trauma was sustained.³ In addition, primate studies documented histologic evidence of axonal shear and strain during experimentally induced acceleration-deceleration mild head injury.^{4,5}

Because the study of concussion does not lend itself to randomized controlled trials, significant strides were made when Barth and colleagues^{3,6} turned to the collegiate sports arena to address the problem of finding adequate control subjects and accounting for the effects of premorbid functioning in the study of concussion. The sporting venue provided a large number of potential participants who were likely to experience mild acceleration-deceleration head trauma. Data from 10 universities showed that athletes who sustained a concussion had neurocognitive deficits at 24 hours and 5 days after concussion but recovered by the 10th day, a recovery curve that has been replicated by other studies.^{7,8} This seminal study set the methodologic standard for baseline and serial neuropsychological testing as the best way to determine the effect of concussion, and this approach of using individual athletes as their own controls and as a model for understanding brain injury in general was termed the Sports as a Laboratory Assessment Model (SLAM).⁹ The SLAM approach helped establish concussion as a sports medicine issue with broader implications for mTBI in other areas (e.g., motor vehicle accidents).

Definition of Concussion

As noted at the outset of this chapter, a widely varying set of terms and defining parameters exist for the topic of sport related concussion (SRC), and this lack of clarity and consistency has hampered the study of concussion to a significant degree. The two most common definitions used in sports concussion research are from the American Academy of Neurology (AAN) Practice Parameters and the Concussion in Sports Group (CISG);

Abstract

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Keywords

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return to play
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neuroscience

commonly referred to as the Vienna, Prague, Zurich, and Berlin Conferences). In 1997 the AAN defined concussion as an altered mental state that may or may not include LOC.¹⁰ Four years later, the CISG group proposed that concussion is a “complex pathophysiologic process affecting the brain, induced by traumatic biomechanical forces,” and included common features of concussion that incorporate clinical, pathologic, and biomechanical constructs to supplement the definition.¹¹ This definition was updated in 2017 to state “Sport related concussion is a traumatic brain injury induced by biomechanical forces,” with common features including “1. SRC may be caused either by a direct blow to the head, face, neck, or elsewhere on the body with an ‘impulsive’ force transmitted to the head; 2. SRC typically results in the rapid onset of short-lived impairment of neurologic function that resolves spontaneously. However, in some cases, signs and symptoms evolve over a number of minutes to hours; 3. SRC may result in neuropathologic changes, but the acute clinical signs and symptoms largely reflect a functional disturbance rather than a structural injury and, as such, no abnormality is seen on standard structural neuroimaging studies; 4. SRC results in a range of clinical signs and symptoms that may not involve loss of consciousness. Resolution of the clinical and cognitive features typically follows a sequential course. However, in some cases symptoms may be prolonged.”¹²

This definition largely reaffirms that established by the prior Consensus Conferences in Zurich^{11,13} and reflects the current trend toward emphasizing the biomechanical force, time course of symptoms, and the functional nature of the injury, as opposed to the structural emphasis used in more severe forms of TBI.

Epidemiology

The Centers for Disease Control and Prevention estimates that approximately 1.6 to 3.0 million sports-related concussions among high school students will be reported each year in the United States, and during the past decade, emergency department visits for sports concussions among children and adolescents increased by 60%.^{14–16} Such estimates likely underestimate the actual occurrence, given that many leagues lack the medical oversight to identify concussions.¹⁷ Recent studies suggest that 8.9% of all high school athletic injuries were concussions,¹⁸ and estimates for collegiate sports range from 5% to 18%, with more than 52,000 concussions reported between the academic years beginning in 2009–2013.^{19,20} The incidence of sports concussion is expected to continue to rise in part because of the increase in sports participation by both male and female athletes at the collegiate and high school levels, but also because of increased knowledge, detection, and reporting of the injury, including the adoption of sports injury surveillance systems.²¹

Pathophysiology

A full discussion of the pathophysiology underlying concussion is beyond the scope of this chapter. The interested reader is referred to the work of Giza and Hovda for a thorough understanding of the topic.^{22,23} In brief, concussion occurs as a result of linear and rotational accelerations and decelerations to the brain and is thought to cause a multifactorial neurometabolic cascade of physiologic changes. This process begins with cell body expulsion

of sodium and potassium into the intracellular space and an influx of calcium, as well as mild edema and a resultant decrease in cerebral blood flow in the setting of initial hyperglycolysis. As glucose is consumed and blood flow is limited, a state of hypoglycolysis ensues, which slows the neurochemical return to homeostasis. This dysautoregulation in addition to axonal injury, impaired neurotransmission, and protease activation leading to cell death explain how clinical signs of brain dysfunction can be observed in the absence of readily detectable physical damage.²³ This cascade begins within minutes of the injury and can be active for several weeks.²³

Clinical Presentation

When the question is raised as to why concussion in sports represents an important problem that must be identified, the answer lies in the known consequences demonstrated by concussed athletes, who may present with acute, catastrophic, lingering, or long-term sequelae.

Although it is controversial and rare, a potentially catastrophic outcome from concussion termed *second impact syndrome* (SIS) is sometimes manifested. This most severe consequence of concussion was part of the initial impetus for a formalized approach to concussion management and RTP criteria. First described in detail in 1984,²⁴ the syndrome is said to occur when a second concussion is sustained before the symptoms of the first have fully resolved. The second impact may appear minor and may not occur on the same day. The athlete may not initially appear injured but may collapse and lose consciousness shortly after the event, with respiratory failure, coma, and even death possibly ensuing. Underlying the SIS phenomena is the notion that the brain appears to have a decreased threshold for sustaining a concussion of any severity with each subsequent trauma.²⁵ The pathophysiology behind SIS is not fully understood, but it has been suggested that the subclinical edema and increased intracranial pressure from the first concussive impact make the brain successively more vulnerable to the second impact through auto-dysregulation and subsequent vascular engorgement. This condition manifests as malignant cerebral edema and herniation of the uncus or the cerebellar tonsils through the foramen magnum, resulting in brainstem failure.²⁴ However, recent systematic reviews of the literature regarding this syndrome found a dearth of high quality literature, unclear actual mortality rates (though nearly all studies said the rate was “high”), and “severe lack of definitive structure” as to what signs and symptoms are associated with it, leading the authors to conclude “SIS is not a credible, literature supported diagnosis.”^{26,27}

Persistent symptoms (also known as postconcussive syndrome or PCS) may remain after a sports concussion is sustained and reflect a failure of normal clinical recovery.¹² Most concussed athletes demonstrate gradual, spontaneous recovery, generally within 2 to 10 days after injury,^{6,7,11,28} consistent with laboratory animal studies showing short-lived effects of concussion,^{22,23} and with meta-analytic studies demonstrating no evidence of impairment beyond 3 months.^{29,30} However, the fact that some people experience persistent sequelae has been recognized for quite some time. For example, at 3-month follow-up, 34% of an mTBI clinical (nonsports) sample had not yet returned to work, and 24%

demonstrated measurable neurocognitive deficits.³ The Fourth International Conference of Concussion in Sport defined prolonged recovery as symptom persistence greater than 10 to 14 days from injury for adults and greater than 4 weeks in children; however, the definitions used in the literature vary from 10 days to more than 2 months.^{13,31,32} Persistent symptoms can include neurologic symptoms such as headache, dizziness, and nausea; cognitive deficits, such as impaired attention, slowed mental processing, and memory dysfunction; and emotional or psychological disruption, including depression or irritability. It is important to note that the presence of one or more of these symptoms after a concussion is sustained in an athlete is not diagnostic of delayed recovery, since such symptoms are extremely common in a host of other conditions. For example, PCS type symptomatology has been demonstrated in 81% of subjects in a chronic pain sample with no history of concussion.³³ As such, the clinician may find the identification of persistent symptoms difficult. One challenge is that the athlete may present with vague, subjective symptoms that are nonspecific. Issues of effort and secondary gain may also be a contributing factor in some cases. Furthermore, the differential diagnosis often includes mood and anxiety disorders, as well as sleep disruption, which can lead to or mimic PCS symptoms. Neuropsychological assessment can be quite useful in this situation, because objective evaluation can help parse out the factors contributing to the athlete's atypical presentation. Structural imaging (MRI and CT) are of limited value in evaluation of these patients and advanced techniques such as fMRI, diffusion tensor imaging, MR spectroscopy, and quantitative EEG have unclear clinical significance.^{31,34}

In addition to the clinical presentations of SIS and PCS, a third potential clinical presentation known as chronic traumatic encephalopathy (CTE) can occur after multiple concussions.³⁵ This outcome appears to be dose-related long-term response of repeated concussions and subconcussive blows and has been studied in groups of retired football, rugby, and boxing athletes.^{36–39} The combination of repetitive sports head injury–related damage with that of age-related neuronal loss⁴⁰ may result in the development of clinical signs of CTE, often appearing years after the end of the athlete's career.⁴¹ Though the diagnosis of CTE is based on neuropathologic evidence collected postmortem, its clinical course can be characterized by long-term deficits in executive functioning, processing speed, verbal learning, and visual memory; motor impairments, including impaired coordination, spasticity, and parkinsonism; as well as behavioral problems such as emotional dysregulation and disinhibition.^{37,42,43} Referred to as “dementia pugilistica” in early literature,⁴⁴ CTE in its advanced stage can be clinically indistinguishable from other advanced dementias. Neuropathologically, CTE is a progressive tauopathy with gliosis and neurofibrillary tangles, as well as common gross elements of reduced brain weight, enlarged ventricles, volume loss in the corpus callosum, cavum septum pellucidum, and scarring and neuronal loss of the cerebellar tonsils.⁴⁵ The pathology of CTE has been most frequently demonstrated in the brains of professional athletes⁴⁶ and can be widespread throughout the brain given the nonfocal nature of repeated head trauma. Similarly, elevated tau protein levels have been detected in the plasma of both injured and noninjured

contact athletes relative to non-athlete controls with increased tau levels corresponding to delayed return to sport.^{43,47} Although long-term deficits in the form of persistent symptoms can manifest after a single concussive blow, it is generally assumed that a concussed athlete can expect a favorable prognosis if the injury is safely managed. The risk of outcomes such as SIS, persistent symptoms, and CTE can be significantly reduced with proper concussion identification, examination, and intervention.

EXAMINATION

The goal of sports concussion examination is to identify the concussion, assess the injury, determine its effects, and devise appropriate interventions. The traditional approach to this task involved simply gauging the signs and symptoms of the concussion and their resolution over time. Many different classification schemes for concussion exist, and definitions and criteria continue to evolve. Historically, the two most widely used systems for grading concussion were those proposed by the AAN in 1997 and Cantu in 2001. These systems used three grades to represent mild, moderate, and severe concussion based on the clinical presentation of the athlete and placed considerable emphasis on LOC (AAN) or posttraumatic amnesia (PTA; Cantu) as severity indicators. Despite early TBI literature showing that coma duration is an important predictor of outcome,⁴⁸ more recent studies have found that the relationship between LOC and injury outcome in the realm of mTBI and sports concussion is less certain.⁴⁹ Concussion grading systems are now considered obsolete, and an individualized approach to concussion diagnosis and management is favored, consistent with the CISG guidelines.

Although monitoring subjective symptoms is an important aspect of concussion management, it cannot be the sole means of gauging effect and recovery. It is now understood that fewer than half of concussed athletes spontaneously report their symptoms,⁵⁰ citing doubt that the injury was severe, not wanting to leave the game, or general lack of awareness about concussion as reasons for not reporting. For these and other reasons, the modern model for concussion examination, outlined by the CISG Conferences, emphasizes an individualized approach that incorporates neurocognitive evaluation. Several computer-administered measures (outlined later) are validated for the assessment of concussion sequelae. The standard practice has evolved to include such standardized tools in both preseason and postinjury assessment. Current guidelines recommend that athletes not resume a gradual progression of activity until their concussion symptoms have remitted. Thus neurocognitive testing has become a valuable tool in identifying subtle cognitive symptoms and their resolution.

Baseline (Preseason) Assessment

Baseline neuropsychological assessments have been advocated by some authors and increased in popularity with the development of computerized tests in the 1990s. Advocates of preseason testing point out the results of any evaluation vary based on many factors other than the actual injury, including intelligence and cultural background, learning disabilities, and previous concussion history.⁵¹ This situation applies to neurologic findings as well; for example, up to 20% of athletes normally experience

a headache while in the midst of competition,⁵² and approximately 3% of the healthy population exhibit pupillary asymmetry.⁵³ Erroneous conclusions about the cognitive and neurologic state of an athlete may be drawn without an awareness of these baseline characteristics. However, other authors have shown these tests may have poor test-retest reliability, vary between group and individual administration or alternate forms, suffer from measurement errors, and may not offer an improvement over postinjury testing alone (when compared with normative data as opposed to individualized preseason baseline).^{54–56} Routine mandatory preseason baseline neurocognitive testing is not recommended by either the Berlin conference or the American Medical Society of Sports Medicine guidelines.^{12,57}

Neurocognitive baselines may be obtained for athletes with the use of computerized methods in an individual or group setting with a trained examiner. Several tools are available for the specific purpose of assessing concussion severity and resolution. These tools include the Automated Neuropsychological Assessment Metrics⁵⁸; the Immediate Post-concussion Assessment and Cognitive Testing (ImPACT Applications Inc., Pittsburgh, PA)^{8,59}; the Concussion Resolution Index (HeadMinder Inc., New York, NY)⁶⁰; and the Computerized Cognitive Assessment Tool (Axon Sports [formerly CogSport], Wausau, WI).⁶¹ The Automated Neuropsychological Assessment Metrics measures processing speed, resistance to interference, and working memory. The Immediate Post-concussion Assessment and Cognitive Testing tool was developed specifically for athletes, and assesses reaction time and a range of attentional and memory skills, accompanied by a self-report postconcussion scale. The Concussion Resolution Index is a web-based set of cognitive tasks that measure simple and complex reaction time, attention, memory, and cognitive processing speed. Axon is also an Internet-based system and includes eight tasks designed to sample a range of cognitive functions.

Sideline Assessment

The first step in the assessment of concussion is to recognize that a concussion may have occurred. It is important to remember that signs and symptoms may be delayed or rapidly changing in the acute phase, and injured athletes may exhibit no obvious indications that they are concussed. Therefore sideline assessment is warranted for any athlete who has received a significant blow to the head or an athlete who does not appear to “be him or herself” in response to a lesser degree of contact.⁶² This assessment involves evaluation of the athlete both on the field, or more commonly on the sideline, as well as in a controlled environment such as the locker room, and usually is conducted by a physician or a certified athletic trainer with expertise in concussion evaluation and management.

An obvious goal of the sideline assessment is to determine if the athlete has signs that signal the presence of severe intracranial or spinal cord injury and whether transport to a medical facility is required.⁶³ Once this determination is made, any individual should be removed from the field for further assessment. Classic signs such as LOC, retrograde amnesia, or frank neurologic deficit may be helpful but are absent in a minority of injured athletes.⁶⁴

In general, the sideline evaluation should assess for signs of confusion, loss of balance, headache, dizziness, and slowed responding, in addition to informal cranial nerve, motor, sensory, and reflex testing⁶⁵ and, importantly, cognitive screening. For athletes with no obvious acute symptoms, physical maneuvers such as push-ups or jumping jacks can help determine whether concussive signs or symptoms develop with exertion and resultant increased intracranial pressure.⁶³ Sideline assessment should also incorporate a mental status examination that targets orientation, concentration, and memory. Notably, standard orientation questions (time, place, person) have been shown to be unreliable following a sports injury relative to memory assessment.^{12,66} Concentration may be gauged by tasks such as repeating strings of three to five digits both forward and backward, serial subtraction of numbers, or repeating a well-known verbal sequence such as months of the year or days of the week backward. Memory can be informally and qualitatively assessed by recall of three items at different intervals after the injury or repeating assignments of a number of previous plays.^{49,65}

Although these informal methods of sideline assessment are helpful, a standardized approach is preferable.⁶⁷ A number of standardized methods for determining the severity of a concussion and the initial symptoms have been developed for use in the sideline evaluation of athletes. The Concussion Recognition Tool 5 (CRT5) was developed by the CISG as a pocket guide for nonmedically trained personnel to identify SRC as well as guide removal from play decisions.⁶⁸ The Standardized Assessment of Concussion (SAC) is a 5-minute measure administered by a trained physician or athletic trainer and includes five orientation questions, a five-word learning test, reciting digits backward, reciting the months of the year in reverse, and delayed word recall.⁶⁹ The SAC provides a 30-point composite score to gauge neurocognitive function and also includes a standard neurologic screening, exertional maneuvers, and means of assessing LOC and PTA. Normative data for comparison are also available.⁷⁰ The SAC is available in paper form and a pocket-card format, and has been adapted for use with smartphone technology. The Balance Error Scoring System is an objective measure of postural stability for use on the sideline.⁷¹ The 5-minute procedure requires the injured athlete to maintain three stances (double, single, and tandem) for 20 seconds with the eyes closed and both hands on his or her hips. Scoring is based on six types of errors over six test trials. The Standardized Concussion Assessment Tool (SCAT5) is another standardized tool that incorporates elements of the aforementioned SAC and a graded rating of common physical and cognitive signs and symptoms.⁷² Recently revised to the fifth version, the test battery takes 10 minutes to administer and updates the widely used SCAT3. This test has also been developed into a pediatric version (Child SCAT5) for use in children age 5 to 12.⁷³ The Berlin Conference recommended use of the SCAT5 and Child SCAT5 for sideline concussion assessment.¹² The tool is easily accessible online and may be freely copied for distribution to individuals, teams, groups, and organizations.

After a sideline evaluation, any athlete considered to have sustained a concussion should be removed from the sporting environment and undergo a multimodal medical evaluation with a more detailed neuropsychological assessment. The concussed

player should not be allowed to return to play (RTP) the day of injury. Serial exams should be conducted over the first few hours after injury to monitor for signs of deterioration. Finally, to provide an objective basis for detecting the effects of the concussion, the athlete should undergo postinjury neurocognitive assessment. This brief repeat evaluation can occur within 24 to 48 hours after the suspected concussion if the athlete appears to be symptom free. This aspect of concussion management may provide increased sensitivity for identifying cognitive impairment resulting from concussion when physical symptoms are limited or resolved.⁷⁴ However, conducting the brief repeat evaluation 7 days after the injury has been found to provide more unique information regarding impairment than routine assessment conducted relatively close in time to the sideline assessment.⁷⁵

Comprehensive Assessment

In the case of a single uncomplicated concussion or subconcussive event, it is generally anticipated that the athlete will experience full resolution of symptoms within the 2- to 14-day natural recovery curve. In cases with an atypical presentation, a more comprehensive evaluation may be necessary. Athletes with preexisting or comorbid conditions (e.g., learning or attention disorders, depression, or substance abuse), those who report or demonstrate a slower than expected recovery, or those who have sustained multiple previous concussions should be considered for expanded assessment. In these situations, a referral for comprehensive outpatient neuropsychological evaluation is indicated, which includes a clinical interview, record review, and objective testing of aptitude and cognitive functioning, and provides individualized recommendations for the athlete and relevant treatment providers. In addition, further medical assessment may be warranted for atypical concussion cases (e.g., neuroophthalmology).

Imaging

Acute evaluation of concussion may incorporate at least one form of brain imaging. Computed tomography is an appropriate choice to rule out the presence of intracranial hemorrhage and skull fracture, and is the imaging option of choice for concussion evaluation in most emergency departments. Magnetic resonance imaging (MRI) with an angiogram can be obtained if carotid dissection or stroke is suspected, and MRI studies may also identify a diffuse injury such as axonal shearing.

It is important to emphasize that most concussions are unlikely to produce observable signs upon testing with these traditional radiologic techniques. Advanced diagnostics such as functional MRI, diffusion tensor imaging, MR spectroscopy, fluid biomarkers, and genetic testing are, at present, only practical and widely available in the research setting and are of uncertain clinical utility.¹² Although structural neuroimaging continues to rule out the presence of serious pathology that would be beyond that of a mild concussive injury, brain imaging is less relevant in the context of typical concussion management.

MANAGEMENT

The cornerstone of acute management of concussion is both physical and cognitive rest. It is theorized that rest may aid recovery by reducing immediate postconcussive symptoms and

decreasing brain metabolic demand.¹² However, there is likely no benefit to continuing strict prescribed rest beyond the acute phase (1 to 2 days) after injury, and most athletes should begin a gradual resumption of activity as tolerated, followed by a graduated protocol to RTP.^{76–78}

An athlete who experiences one concussive event is three times more likely to sustain another concussion in the same season, usually within 10 days of the first injury.⁷⁹ As previously outlined, the athlete is at heightened risk for severe sequelae of repeated concussions, particularly during the acute stage when the brain is more susceptible to subsequent insult. The prolonged recovery of multiply concussed athletes has been demonstrated with lingering cognitive and gait stability deficits.^{25,80} Players who sustain a concussion are more likely to experience LOC with the next injury.^{25,81} Furthermore, the force threshold for concussive injury may be far less for a second impact⁷⁹; therefore subconcussive events must be considered as possible risk factors when discussing concussion outcome and management. A subconcussive blow is an apparent brain insult with insufficient force to cause the hallmark symptoms of concussion.⁸² These more minor events conceivably occur at a high rate during both competitive play and practice. Investigation with use of the Head Impact Telemetry System to record the frequency, magnitude, and location of head impacts found that a group of college football players sustained on average 1000 subconcussive impacts over the course of a single season.⁸³ The notion that repetitive blows may cause equivalent, if not greater, damage than a single mild concussion has been suggested previously.⁸⁴

It is possible that the number of concussions sustained plays a more important role in long-term outcomes than the severity of the concussions. The animal experimental literature has shown that two or more concussive blows produce greater deficits than predicted by single blows and that repeated blows can accelerate the deposition of β -amyloid.⁸⁵ The heightened risks of repeat concussive injury have also been extended to human studies.^{79,86} Contact athletes with a history of multiple concussions have an increased risk of depression and cognitive deficits later in life (though, notably, the majority of former players assess their functional status as normal).⁸⁷ What is unknown is “how many is too many,” and this number will vary for individual athletes.

Different groups and individuals are more susceptible than others, and the nature of their symptoms and deficits may vary. For example, younger athletes appear to be particularly vulnerable to negative outcomes, with greater susceptibility to SIS and longer recoveries.⁸⁸ Compared with their college counterparts, high school athletes may be more vulnerable to concussion, exhibit a more protracted recovery, and have longer lasting cognitive deficits.^{8,89} However, the literature is insufficient at present to distinguish between the presentation, management, or prognosis between children, adolescents, and adults. Differences by gender may also exist, with some evidence that females experience a higher incidence of persistent symptoms compared with their male counterparts, although this literature is still equivocal.^{90,91}

Given the well-documented risks involved in returning an athlete to play, the potential for delayed onset of symptoms; variable symptom reporting; the nonlinear, individualized nature of recovery; and the known long-term risks for such consequences as CTE, the accepted standard for concussion management is

that concussed athletes are held from competition until they are completely asymptomatic at rest and during exertion (which involves either physical or cognitive activities). When the previously outlined examination process is followed, the clinician is equipped with appropriate data with which to approach an RTP determination.

Return to Play

The primary focus of concussion management revolves around the appropriate implementation of RTP guidelines. Historically many different classification systems have been used to guide RTP decisions. The AAN published multiple guidelines and classification systems for concussion management from 1990 to 2005, all of which were based on clinical experience and expert opinion. The Cantu system of concussion grading was also revised to guide RTP decisions, and the Colorado Medical Society is another frequently referenced model for RTP.^{92,93} These models added much to the field of sports concussion but were criticized for lack of empirical basis and standardization, and for being overly restrictive.

Newer approaches have replaced the use of grading systems of acute symptoms with more emphasis on the individual's path of symptom resolution rather than preconceived timelines. This process has been guided by recommendations of the CISG, which produced a series of consensus documents that evolved from the first summary statement in 2001 (Vienna Conference) to a second statement in 2004 (Prague Conference), the third and fourth iterations following the Zurich Conferences in 2009¹¹ and 2013,¹³ and the Berlin conference in 2016.¹²

The consensus models a graduated recovery and rehabilitation strategy. The current Berlin protocol (Table 126.1), like its predecessor, calls for immediate removal from practice or play after a suspected concussion, standardized sideline assessment, and no RTP on the day of injury. The protocol then follows a stepwise process that progresses through six graduated steps that take 24 hours each. If any postconcussion symptoms occur within a step, the athlete is returned to the previous asymptomatic level for another 24 hours until resuming the graded process. It should be noted that cognitive rest and restriction from cognitive exertion (e.g., schoolwork, video gaming, and texting) is as equally important as physical rest and restriction from physical exertion during the course of recovery. Clinicians involved in the care of

athletes should be well versed in the guidelines and should engage in a dynamic approach to RTP that takes into account the individual's history with careful attention to age, number of previous concussions, severity of concussion, and comorbidities such as other health issues, mood or psychiatric history, and substance use. The Berlin consensus statement recommends that children and adolescents not return to sports until they have successfully returned to academics with the caveat that early introduction of symptom-limited activity is appropriate.¹² The 2013 AAN guidelines suggest a conservative approach to RTP for children and further recommend no RTP for any concussed athlete until he or she is assessed by a licensed health care professional trained in concussion management.⁹⁴

Retiring an Athlete

One of the most challenging aspects of sports concussion management is the issue of when to retire an athlete from play. In the face of limited empirical data on the topic and the great variability in both immediate symptoms and long-term effects that individual athletes demonstrate after sustaining single or multiple concussions, no clearly defined method exists to guide the clinician or team.⁹⁵ At the collegiate and professional level, the financial, social, and legal implications also weigh into such decisions. However, several important factors clearly must be taken into consideration to make a reasoned decision, including age, sex, medical and concussion history, sport, position and level of play, and concussion specific symptoms and testing results.⁹⁶ When an athlete begins to demonstrate a longer period of recovery from symptoms, experiences concussive symptoms from forces that are increasingly mild, or does not return to baseline neurocognitive functioning, then it may be time for the athlete to stop playing.⁹⁷ In addition, the athlete's functioning in multiple settings should be considered; for example, problems at school or home or behavioral problems can signal the need to consider retirement from play. A particularly conservative approach is recommended with children, where the decision must be based solely on the future health of the patient.⁹⁶

INTERVENTIONS

Sport concussion interventions are targeted in three main areas. The most critical means of intervention is education. Concerted

TABLE 126.1 Concussion in Sports Group Graduated Return-to-Sport Protocol

Rehabilitation Stage	Functional Exercise at Each Stage of Rehabilitation	Objective of Each Stage
1. Symptom-limited activity	Daily activities that do not provoke symptoms	Gradual reintroduction of work/school activities
2. Light aerobic exercise	Walking, swimming, or stationary cycling at slow to medium pace No resistance training	Increase heart rate
3. Sport-specific exercise	Running or skating drills; no head impact activities	Add movement
4. Noncontact training drills	Harder training drills (e.g., passing drills) May start progressive resistance training	Exercise, coordination, and increased thinking
5. Full contact practice	After medical clearance, participate in normal training activities	Restore confidence and assess functional skills by coaching staff
6. Return to play	Normal game play	

From McCrory P, Meeuwisse W, Dvorak J, et al. Consensus statement on concussion in sport—the 5th International Conference on Concussion in Sport held in Berlin, October 2016. *Br J Sports Med*. 2017;bjsports-2017-097699. doi:10.1136/bjsports-2017-097699..

efforts must be made to improve concussion awareness for athletes, their parents, coaches, athletic trainers, and school and league administrators. A basic understanding of what constitutes a concussion, the signs to look for, the importance of removing the athlete from further concussion risk, and the need to obtain an appropriate examination will go a long way toward improving outcomes. The need for education is highlighted by actions such as the passage of the Lystedt Law, named for Zackery Lystedt, who sustained a TBI when he returned to a middle school football game after experiencing a concussion. The legislation, passed in 2009, aims to protect athletes by mandating that athletes, parents, and coaches be educated about the dangers of concussions each year, that a player suspected of having a concussion be removed from play, and that a licensed health care professional clear the athlete for RTP.⁹⁸ “Lystedt laws” are being enacted nationwide to outline educational requirements and the need for clearance by a qualified professional any time a concussion is suspected.

A second area of intervention is, of course, prevention. The data on protective equipment are somewhat mixed with regard to whether headgear and helmets actually reduce concussion incidence. It has been suggested that such equipment has little ability to stop concussion, because helmets are not able to prevent deceleration and rotation forces on the brain.⁹⁹ The most powerful means of prevention may be rule changes within the sport, and a clear precedent exists for organizational changes. For example, the NFL made the tackling method of using one’s head or helmet to hit another player (“spearing”) an illegal technique in an effort to curb serious injury. Likewise, disallowing body checking in ice hockey for children under 13 has resulted in a significant reduction in concussion rate among these players.^{100,101} Attention should be directed to both game time and practice activity, since the incidence of concussion in each phase is roughly similar.²⁰ Rules such as limiting helmet-to-helmet contact by enacting penalties or other consequences would have beneficial effects for athlete protection and brain health. Furthermore, fair play and the distinction between aggressive play and violence should be emphasized within sporting organizations.

CONCLUSION

The neuroscience and clinical study of sports concussion remains a developing area that has increasingly become the focus of both laypeople and researchers alike in light of the knowledge that concussions can exert meaningful effects on the athlete both in the short and long term. Given the number of variables that can influence concussion symptoms and length of recovery, individualized multimodal clinical management of concussed athletes guided by a clearly defined concussion evaluation process is paramount. The challenge for the future is to delineate factors that contribute to individual susceptibility to concussive blows, the variables that influence prognosis, and methods of preventing catastrophic, prolonged, and long-term outcomes, as well as to increase the awareness of sport concussion in players, their families, sporting organizations, treatment providers, and the general public.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Belanger HG, Vanderploeg RD. The neuropsychological impact of sports-related concussion: a meta-analysis. *J Int Neuropsychol Soc*. 2005;11(4):345–357.

Level of Evidence:

II

Summary:

This meta-analysis of 21 studies involving 790 cases of concussion and 2014 control cases was conducted to determine the impact of sports-related concussion across six cognitive domains. Results provide compelling evidence that sports-related concussion has no significant effect on neuropsychological function by 7 to 10 days after injury in the athletic population at large, although long-term participation in sports involving head contact may be associated with small, adverse sequelae.

Citation:

Echemendia RJ, Cantu RC, et al. Return to play following brain injury. In: Lovell M, Echemendia R, Collins M, eds. *Traumatic Brain Injury in Sports: an International Neuropsychological Perspective*. Lisse, The Netherlands: Swets and Zeitlinger; 2004.

Level of Evidence:

V

Summary:

The authors propose a model for return-to-play decision-making that takes into account the complexity of multiple factors and variables. Informed decision-making involves consideration of data from many different sources.

Citation:

Giza CC, Hovda DA. The neurometabolic cascade of concussion. *J Athl Train*. 2001;36(3):228–235.

Level of Evidence:

II

Summary:

The authors use more than 100 articles from both basic science and clinical literature to review the underlying pathophysiologic processes of concussive brain injury and relate these neurometabolic changes to clinical sports-related issues such as injury to the developing brain, overuse injury, and repeated concussion.

Citation:

Gysland SM, Mihalik JP, Register-Mihalik JK, et al. The relationship between subconcussive impacts and concussion history on clinical measures of neurologic function in collegiate football players. *Ann Biomed Eng*. 2011;40(1):14–22.

Level of Evidence:

II

Summary:

This study is the first to investigate the relationship between repetitive subconcussive head impacts and clinical measures of neurologic impairment while monitoring sustained impacts throughout the course of a single collegiate football season. The methodologies presented could help establish a foundation for future longitudinal study aimed at answering questions about the

possible relationship of recurrent subclinical blows and later depression, early-onset dementia, and chronic traumatic encephalopathy.

Citation:

McCrory P, Meeuwisse W, Dvorak J, et al. Consensus statement on concussion in sport—the 5th International Conference on Concussion in Sport held in Berlin, October 2016. *Br J Sports Med.* 2017;bjsports-2017-097699. doi:10.1136/bjsports-2017-097699.

Level of Evidence:

V

Summary:

These guidelines improved on those previously presented by the same conference in Vienna, Prague, and Zurich, and constitute the most widely used protocol for our current understanding of and approach to sport concussion.

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